

NULA DeepSleep ESP32-S3

The NULA DeepSleep ESP32-S3 is a low-power development board with 7 μ A deep sleep current, ideal for battery-powered IoT applications.

The **NULA DeepSleep ESP32-S3** is a powerful development board engineered for ultra-low-power IoT applications. Built around the **ESP32-S3-WROOM-1-N8R8** module, it combines a dual-core processor with 8 MB of Flash and 8 MB of PSRAM, providing ample resources for complex projects. The board's standout feature is its exceptionally low deep sleep current of just 7 μ A, making it an ideal choice for battery-powered devices where power efficiency is crucial. With integrated **Wi-Fi** and **Bluetooth** LE connectivity, the NULA DeepSleep is the perfect center-stage for your embedded projects. Due to its ample connectivity, it excels in general use applications as well, such as display driving and wireless communicating.

It features an integrated **PCF85063A** real-time clock (RTC) for precise timekeeping, even when the main processor is in deep sleep. Power management is handled by an onboard **TP4056M Li-Ion/Li-Po** battery charger, allowing for battery integration and charging. The board's design ensures stable performance and provides all the necessary components for a long-lasting IoT solution right out of the box. A **USB-C to USB-C** cable is included in the package, so you can start programming immediately without any additional accessories.

Power your next project with the NULA DeepSleep ESP32-S3 if you are building remote environmental sensors, asset trackers, or low-power wireless nodes. This board provides the performance and efficiency you need with its breadboard-friendly layout that makes prototyping quick and easy, while the rich feature set supports a wide range of applications.

MCU

Microcontroller	ESP32-S3
Mcu Clock (MHz)	240 MHz
Sram (KB)	512 KB
Flash (MB)	4 MB

Connectivity

Wifi	Wi-Fi 4 (802.11n)
Bluetooth	Bluetooth Classic + BLE
Wireless Module	ESP32-S3-WROOM-1(N8R8)

Interface

Qwiic Compatible	Yes
Breadboard Compatible	Yes

Power

Battery Connector	JST 2-pin connector
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Onboard Battery Charger	Yes
Supply Voltage	3.3 V or 5 V
Logic Level	3.3 V
Sleep current	16 μ A

Other

Dev Environments	Arduino IDE, MicroPython, ESP-IDF
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Variants

333352	No Headers
333369	Male Pin Headers

What's in the box

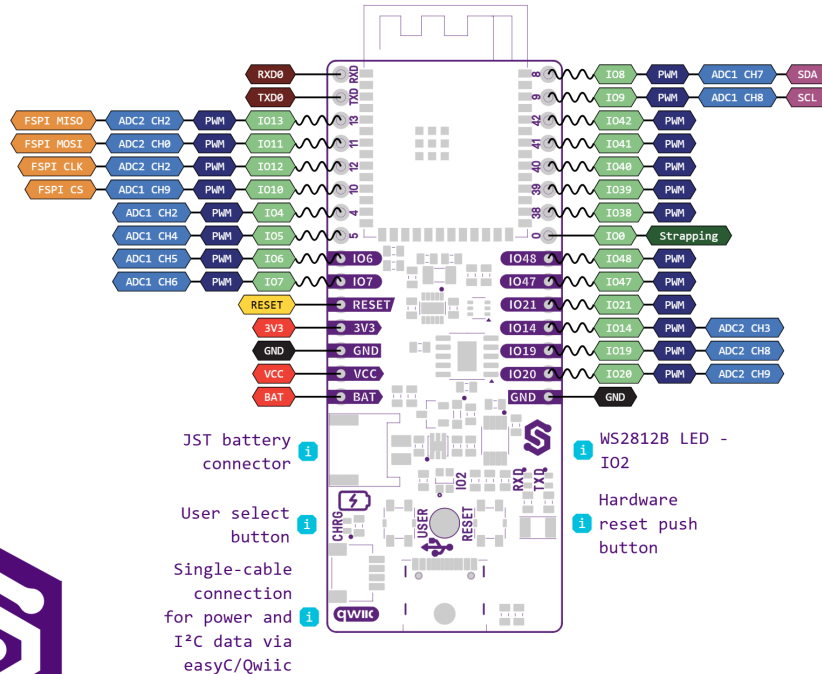
1× NULA DeepSleep ESP32-S3 (main product)	
1× Header pins (unsoldered)	Header pins included for all breadboard-compatible pins
1× USB-C cable	

Pinout

SOLDERED NULA ESP32-S3 LOW-POWER - Pinout

Full product documentation:
solde.red/333352

JP1 (NC) - Disconnect WS2812B LED power supply



v1.0

- Pin description
- Power supply pin
- Ground
- Control
- GPIO
- ADC pin
- DAC pin
- SPI comms
- UART comms
- I2C comms
- PWM comms
- Other
- Port name
- General info
- Warning!



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Typical applications

NULA DeepSleep suits battery-powered projects that spend most of their time asleep and wake briefly to take a reading or send data. Typical uses include a remote temperature logger, a mailbox or door sensor that reports over Wi-Fi, or a battery-powered prototype that needs qwiic sensors without extra wiring. The onboard battery charger lets you run it directly from a LiPo cell and recharge over USB-C. Because it supports the Arduino IDE, MicroPython, and ESP-IDF, you can start with simple example code and move to lower-level control later if your project needs longer battery life.

Resources and downloads

Hardware Details

<https://docs.soldered.com/nula-deepsleep-esp32-s3/hardware-details/> (WEB)

Pinout

https://cms.soldered.com/products/333352/media/333352_pinout-png_b3cefc.png (WEB)

Full Documentation

<https://docs.soldered.com/deepsleep-esp32-s3/overview/> (DOCS)